Experiment 13

Limitations of Low-Code Development Platforms Using Zoho Creator

Aim

To study and analyze the limitations of low-code development platforms using Zoho Creator, focusing on customization, performance, scalability, integration, security, cost and platform dependency, and to identify possible solutions for overcoming them in the Transport Management System.

Procedure

1. Study the features of a low-code platform and the kinds of applications it is meant to build.
2. Open the Transport Management System in Zoho Creator and examine its dashboard, forms, reports and workflows.
3. Identify where customization is restricted, for example in the layout and components available on a page.
4. Review the performance, scalability and integration constraints that arise as records, workflows and users increase.
5. Examine the workflow and other automation options to see how they behave and where they are limited.
6. Check the subscription plan comparison and the features locked in the current plan to understand cost and usage limits.
7. Analyze security, offline access and advanced technology limitations.
8. Record the limitations found and note a suitable solution for each one.

Limitations of Low-Code Development Platforms

2.1 Limited Customization

Predefined layouts and components restrict full control over the user interface. For example, the dashboard can only use the available KPI cards and chart types.

Solution: Use custom pages, supported UI components and scripting where available.

2.2 Performance Limitations

Performance may drop as vehicle, trip and passenger records, workflows, reports and calculations increase; complex scripts and repeated queries slow the system.

Solution: Optimize Deluge scripts, filter records before processing, fetch only required data and minimize unnecessary workflow runs.

2.3 Scalability Challenges

Platforms may limit data storage, workflow executions, API calls and concurrent usage as the fleet size or number of trips grows.

Solution: Review platform limits, optimize design, monitor usage and consider higher-capacity plans or external services.

2.4 Integration Limitations

Not every external service, database or legacy system is directly supported; for example a third-party GPS tracking system.

Solution: Use supported APIs, webhooks and connectors while ensuring secure data exchange.

2.5 Platform Dependency (Vendor Lock-in)

Deluge scripts and configurations are specific to Zoho Creator, so migrating to another platform requires significant effort.

Solution: Maintain documentation, back up data and evaluate migration needs before choosing a platform.

2.6 Cost and Usage Limitations

Plans limit users, storage, automation and advanced features such as payment workflows; costs rise as usage grows.

Solution: Review plan limits, estimate future needs and choose a plan based on usage and budget.

2.7 Security and Compliance Challenges

Wrong permission settings can expose driver licence details, mobile numbers or passenger contact details to unauthorized users.

Solution: Implement role-based access control, restrict confidential fields, validate inputs and review audit logs.

2.8 Limited Offline Access

Cloud-based applications depend on connectivity; network interruptions can affect trip scheduling and workflow execution at the depot.

Solution: Evaluate offline capabilities and define alternative procedures for connectivity issues.

2.9 Limitations in Advanced Technologies

Specialized algorithms such as route optimization or predictive maintenance models may need extra development.

Solution: Use supported integrations and external AI services while considering security, performance and maintenance.

Observations

While the Transport Management System was built in Zoho Creator, the forms, reports, workflows and the dashboard were created quickly with very little coding. The limits became visible when the design went beyond the sample data and the basic features.

The dashboard can use only the available Page Builder elements, the payment workflow and higher usage limits need a paid plan, and a GPS tracking system needs API and connection setup. Role-based permissions were also needed so that driver and passenger details stay visible only to authorised staff.

Summary of Limitations Observed in This Project

Limitation	Example in the Transport Management System	Solution
Limited customization	Dashboard uses only the available KPI cards and charts	Custom pages and scripting
Performance	Slows with many records and repeated queries	Criteria-based fetch, optimized scripts
Scalability	Limits on storage, API calls and workflow runs	Monitor usage; higher plan
Integration	Third-party GPS system needs API setup	APIs, webhooks, connectors
Vendor lock-in	Deluge and configuration are Zoho-specific	Documentation and backups
Cost	Payments workflow and higher limits need a paid plan	Choose the plan by usage
Security	Wrong permissions can expose driver details	Role-based access and audit logs
Offline access	Trip scheduling needs connectivity	Manual fallback procedure
Advanced technology	Route optimization needs extra development	External AI services

Result

The limitations of low-code development platforms were studied using Zoho Creator and a possible solution was identified for each limitation.

Conclusion

Low-code platforms such as Zoho Creator simplify application development by reducing coding effort, accelerating development and providing built-in tools for forms, reports, workflows and dashboards. However, they have limitations related to customization, performance, scalability, integration, platform dependency, cost, security, offline access and advanced technologies. These must be considered during application planning and implementation.